



Official TCC Course Syllabus

Discipline Prefix: MTH	Course Number: 267	Course Title: Ordinary Differential Equations	
	Course Section: OL1C		
Credit Hours: 3	Lecture Hours: 3	Clinical Hours:0	Lab Hours: 0
Contact Hours: 3	Studio Hours:	Repeatable for Credit: No XX	
Semester: Fall 2023 Class Start Date:08/21/2023 Class End Date: 12/05/2023		Meeting Days/Time/Location: Asynchronous Online Building VIRT Room VIRT - Virtual Location	

Instructor Information

Name: Professor Pamela Dale

Office Location: Virtual

Office Hours: Via ZOOM or phone conference by appointment (evenings and weekends included)

Contact Information:

e-mail: pdale@tcc.edu office phone: 757-822-5001(e-mail or OHM message preferred)

Canvas site: <http://learn.vccs.edu>

Course Description

Introduces ordinary differential equations. Includes first order differential equations, second and higher order ordinary differential equations with applications and numerical methods.

Prerequisites and/or Co-requisites

Prerequisites - MTH 174 or MTH 264 with a grade of C or better. Corequisites - None

General Education Core Competencies Supported by this Course

After completion of this course, students will be able to demonstrate proficiency in Quantitative Literacy:

- Quantitative Literacy is the ability to perform accurate calculations, interpret quantitative information, apply and analyze relevant numerical data, and use results to support conclusions. Degree graduates will calculate, interpret, and use numerical and quantitative information in a variety of settings.

Required Course Texts and Supplementary Materials: Reliable access to a computer with internet service.

Item #1 (Electronic Versions): OER material is provided for free in PDF format:

Trench, William F., "Elementary Differential Equations" (2013). Books and Monographs. Book 8. <http://digitalcommons.trinity.edu/mono/8>

Item #2: On-line Math Program: This course uses Lumen Learning, on-line management system:

<https://ohm.lumenlearning.com/>

- ISBN/Generic Code is: 978-1-64087-163-2
- Course ID: 75034 and Key is Mathtime!** (Last symbol is an exclamation mark)

Course Learning Outcomes

- Identify and solve first-order differential equations and applications that include the following: A. Separable B. Exact C. Linear D. Bernoulli
- Identify and solve linear differential equations of order greater than one and applications using methods that include the following: A. Form of Known Solution B. Undetermined Coefficients C. Variation of Parameters
- Use Laplace transforms to solve differential equations.
- Calculate numerical solutions to differential equations using the following: A. The Euler Method B. Three Term Taylor Method C. Runge-Kutta Method

Topics Covered in the Course

- First-Order Differential Equations: Classify a differential equation as linear or nonlinear. Sketch a directional field for an arbitrary first-order differential equation. Determine the order, linearity or nonlinearity, of a differential equation. Solve first order linear differential equation. Solve separable differential equations. Solve initial value problems.
- Numerical Approximations: Use the Euler or tangent line method to find an approximate solution to a linear differential equation.
- Higher-Order Differential Equations: Solve second order homogeneous linear differential equations with constant coefficients including those with complex roots and real roots. Determine the Fundamental solution set for a linear homogeneous equation. Calculate the Wronskian. Use the method of Reduction of order. Solve nonhomogeneous differential equations using the method of undetermined coefficients. Solve nonhomogeneous differential equations using the method of variation of parameters.
- Laplace Transforms: Use the definition of the Laplace transform to find transforms of simple functions. Find Laplace transforms of derivatives of functions whose transforms are known. Find inverse Laplace transforms of various functions. Use Laplace transforms to solve ODEs.
- Numerical Methods for Ordinary Differential Equations
- Applications of Differential Equations: Springs-Mass-Damper, Electrical Circuits, Mixing Problems. Solve applications of differential equations as applied to Newton's Law of cooling, population dynamics, mixing problems, and radioactive decay. (1st order). Solve springs-mass-damper, electrical circuits, and/or mixing problems (2nd order). Solve application problems involving external inputs (non-homogeneous problems)

Description of Assignments/Assessments

Course assignments and assessments will consist of homework assignments, one conceptual assessment and two proctored exams.

Course Schedule

The course schedule may change at the discretion of the instructor; however, students will be notified in writing when any changes/additions are made to the schedule.

Week 1: Tuesday August 22nd through Tuesday August 29th

- Enroll in Lumen OHM Math Program
- "Selfie" Verification and Introduction Assignment
- Basic Concepts
- Directional Fields
- Linear First Order Equations
- Separable Equations

Week 2: Tuesday August 29th through Tuesday September 5th

- Existence and Uniqueness of Solutions of Non-linear Equations
- Bernoulli Equations
- Application: Growth and Decay

Week 3: Tuesday September 5th through Tuesday September 12th

- Exact Functions and Equations
- Integrating Factor
- Newton's Law Cooling/Heating and Mixture

Week 4: Tuesday September 12th through Tuesday September 19th

- **Test #1 (Proctored)** covering material from Weeks 1 through 3

May choose one of the three ZOOM testing sessions with me (Free) or taken at any one of the four TCC testing centers (Free) or another *approved* proctor (i.e. military education officer, another college, etc. usually free), or use the on-line secure proctor RPNOW (fee of \$15).

Details in the Syllabus Folder inside Lumen (Math Program)

Week 5: Tuesday September 19th through Tuesday September 26th

- Second Order Homogeneous Linear Equations with Constant Coefficients, Linearity, and Wronskian
- Higher Order Homogeneous Equations with Constant Coefficients

Week 6: Tuesday September 26th through Tuesday October 3rd

- Method of Undetermined Coefficients

Week 7: Tuesday October 3rd through Tuesday October 10th

- Reduction of Order
- Variation of Parameters

Week 8: Tuesday October 10th through Tuesday October 17th

- Motion, Spring Systems, RL Circuits

****Week 9: Tuesday October 17th through Tuesday October 24th**

- Conceptual Assignment on Second Order

Week 10: Tuesday October 24th through Tuesday October 31st

- Laplace Transformations
- Inverse Laplace Transformation

Week 11: Tuesday October 31st through Tuesday November 7th

- Applying Laplace to solving differential equations
- Unit Step Function

Week 12: Tuesday November 7th through Tuesday November 14th

- Series and application for solving differential equations
- Cauchy-Euler

Week 13: Tuesday November 14th through Tuesday November 21st

- Numerical Methods
- Euler's Method
- Improved Euler's Method
- Runge-Kutta Method

******Week 14/15: Tuesday November 21st through Tuesday December 5th (covers Thanksgiving week)

- **Test #2 (Proctored)** covering material from Unit #3

May choose one of the three ZOOM testing sessions with me (Free) **or** taken at any one of the four TCC testing centers (Free) **or** another *approved* proctor (i.e. military education officer, another college, etc. usually Free), **or** use the on-line secure proctor RPNOW (fee of \$15).

Details in the Syllabus Folder inside Lumen (Math Program)

Late Passes: A total of 3 late passes **for homework** are allotted for illness, out-of-town, internet issues, alien abductions, good powder on slopes, rolling wave sets for surfing, Mars attacks, etc. and are issued at the beginning of the semester.

- Good for one week after the original homework due date.
- One late pass is required for each past due assignment.
- May be applied on homework only. Tests/Exams/Conceptual must be completed on-time.
- Use them sparingly, once they are used....no more given 😊.

Fall 2023 Important College Dates for 15-week Session

- | | |
|--|---------------------------|
| • First day of class | August 21 st |
| • Drop deadline for tuition refund | September 5 th |
| • Last date to withdraw without academic penalty | October 23 rd |
| • Last Day for this class | December 5 th |

Note: *Students who wish to withdraw without academic penalty should contact a counselor to determine the appropriate procedure and date. Withdrawals through completion of sixty percent of a session will result in a W grade. After sixty percent of a session is completed, a withdrawal will result in a grade of F in a credit course or a grade of U in a developmental course, except under mitigating circumstances that must be documented by the instructor and approved by the academic dean.*

While it is the student's responsibility to withdraw from class, due to updated Federal and College attendance policies, a student who is inactive for more than 14 consecutive days prior to October 23rd 2023, will be withdrawn from class with a grade of "W" due to lack of attendance. A student who stops attending will receive a grade of "F". In short, if you stop attending, please officially withdraw prior to October 23rd 2023.

Course Policies

Generative AI (Artificial Intelligence) Policy: Students are expected to complete work without the use of AI including but not limited to ChatGPT, Symbolab, Brilliant, Photomath, Mathway. etc.

Homework Policy: Homework assignments will be posted on <https://ohm.lumenlearning.com/>

Students are expected to complete the assignments by the due dates. Students will need to enroll in the Lumen OHM math program to complete the online assignments.

Test Policy: There will be two timed proctored tests over the course of this semester. Details are located on the Lumen math program in the Syllabus folder.

Grade Policy

The course grade will be determined as follows:

Average of all On-line Math (OHM) Weekly Graded Homework:	40%
TEST #1 (Proctored....reference note below)	25%
TEST #2 (Proctored....reference note below)	25%
Conceptual Assignment	10%

Details for Proctored Test #1 and #2. You may (choose one)

- Choose one of the ZOOM sessions offered (Free..with me)
- Go to one of the four TCC testing centers (Free...but must schedule an appointment).
- Use another *approved* proctor (i.e. military education officer, another testing center...usually Free).
- Utilize the online secure proctor RPNOW, which is the approved live online proctoring service (fee of \$15 reference details in the Syllabus Folder on Lumen)

All grades are recorded on Lumen OHM math program, not on Canvas. Official Final Course Grades are now delivered via the web on SIS.

Graded assignments and exams will be graded numerically and will be converted to a letter grade according to the following scale: 90 - 100 = A, 80 - 89 = B, 70 - 79 = C, 60 - 69 = D, Below 60 = F

Online Course Communication

Student postings, messaging (using the OHM math program is the preferred method), and e-mails will be responded within 24 hours during the week or within 48 hours over weekends and holidays by the instructor.

NOTE: Priority is placed on interaction with online students, and I usually respond the same day; *HOWEVER*, forum postings, messaging, or e-mails sent on the Tuesday night assignments are due, may not be answered until the next day.

Forum Discussion on OHM

Students can receive up to 5 extra credit points toward each test (1 point per question and/or answer) for participating in The Math Forum. Personal questions should be sent via email or message.

Conditions to earn bonus points on math forum:

- Posts must have some mathematical substance, including a specific question(s) i.e., a general “I do not know how to do this” does not count ☺
- Students need to state where in the problem the misunderstanding has occurred and write something about the problem. (Do not just post the problem). Including the type of problem, why you cannot solve it, etc. is needed.
- Credit is awarded for questions AND answers (peer learning is encouraged)!

Attendance/Tardiness Policy:

All students are expected to be present and on time at all scheduled class and laboratory meetings, whether in person or online. Instructors are not required to admit or credit for participation a student who arrives late to the course meeting. A student who adds a class or registers after the first day of classes is counted absent from all class meetings missed, including those prior to registration.

If a student is absent more than 15 percent of scheduled instructional time, attendance may be defined as unsatisfactory. This calculation includes absences occurring during the add/drop period. See also the Withdrawal Policy in this syllabus for more information. Per the college's attendance policy, faculty have the right to develop a more stringent policy as well. Students who do not attend or participate in class by the deadline to drop for tuition refund may be deleted from the course.

In a distance learning course, student attendance is determined by course participation as defined by the faculty. Students who enroll in a course but do not meet federal attendance requirements shall be administratively deleted or withdrawn from the course in accordance with federal requirements. Existing college policies regarding tuition refund shall remain in effect.

The following attendance policies are specific to Calculus Online/Internet

- Allow 8 - 10 dedicated hours a week for this class.
- Students are expected complete assignments (OHM) on a weekly basis.
- A week is considered from 0001 Tuesday through 2359 the following Tuesday.

Late Work/Make-up Exam Policy:

There are 5 late homework passes available to each student at the beginning of the course. A late pass is available for homework assignments only and extend the due date one week after the original homework due date. These passes are available for reasons of illness, family emergencies, out-of-town, loss of internet, etc. No additional passes will be awarded, be sure to use them sparingly.

Classroom Behavior

TCC is committed to maintaining an inclusive environment conducive to carrying out its mission; therefore, all members of the TCC community are expected to demonstrate standards for civility.

- Be moderate in communication. Loud, obscene, argumentative, or threatening communication is disruptive to teaching and learning.
- Resolve any disagreements in a positive, non-combative manner. Request the assistance of college authorities if needed.
- Show respect for the comfort of others in an educational setting by observing acceptable standards for dress and surroundings, whether in the classroom or participating in an online meeting.
- Adhere to the [Tidewater Community College Code of Student Rights and Responsibilities](#).

Statement on Regarding Online Behavior: Proper online etiquette is to be followed.

Electronic Devices Policy: Not applicable for this class.

Disposition of Classes for Emergency Shutdown of the College Does not apply for this online course.

Academic Policies

Students are responsible for being aware of the policies, procedures, and student responsibilities contained within the current edition of the TCC *Catalog* and *Student Handbook*. Students should familiarize themselves with the college's policies regarding misconduct and inclement weather found in the *Student Handbook*.

Withdrawal Policy

Students who wish to withdraw without academic penalty should contact a counselor to determine the appropriate procedure. Withdrawals through completion of 60 percent of a session will result in a W grade. After 60 percent of a session is completed, a withdrawal will result in a grade of F in a credit course or a grade of U in a developmental course, except under mitigating circumstances that must be documented by the

instructor and approved by the academic dean. Dynamic session classes have unique refund and withdrawal dates. Contact a campus Enrollment Services Office for more information or visit the Academic Calendar website (URL provided in Important Websites section).

A student who drops after the last day to withdraw does not receive a 'W'. He/she receives an 'F', in which case there is both an academic and financial penalty. A student who withdraws by the deadline faces a financial penalty, but not an academic penalty.

September 5 2023	Last day to drop for tuition refund from a 15-week course
October 23 2023	Last date to withdraw without academic penalty (receive W)

Academic Integrity

TCC will expect students to demonstrate personal and academic integrity, to be open to new ideas, and to share in a community where individuals from diverse backgrounds and cultures help one another grow intellectually, socially, and personally. TCC expects students to achieve, not just to get by. And while many caring and talented faculty and staff are here to help, students must take responsibility for their own learning. Students should strive for a high level of academic performance and to be responsible, contributing citizens within the college and in outside communities. Above all, TCC wants students to develop a love of learning that will last a lifetime, along with a life-long interest in maintaining emotional and physical wellness.

Student Outcomes Assessment Requirement

Work products submitted by students fulfill course requirements may be used by the college to evaluate its academic programs and general education requirements.

Plagiarism and Academic Misconduct

Academic misconduct includes, but is not limited to, the following actions: cheating on an examination or quiz—either giving or receiving information; copying information from another person for graded assignments; using unauthorized materials during tests; collaboration during examinations; buying, selling or stealing examinations; arranging a substitute for oneself during examinations; substituting for another person, or arranging such a substitution; plagiarism—the intentional or accidental presentation of another’s words or ideas; collusion with another person or persons in submitting work for credit in class or lab, unless such collaboration is approved in advance by the instructor.

Faculty members who have reliable evidence of academic misconduct will (1) investigate the matter, and (2) review the facts of the matter and the proposed penalty with the appropriate academic dean. They may then take one or more of the following actions:

- ☐ Require the work to be accomplished again
- ☐ Give no credit for the test, paper, or exercise
- ☐ Assign a grade of **W** or **F** for the course
- ☐ Refer the matter to the campus Dean for Student Services or designee for possible disciplinary sanction through the college’s disciplinary procedure

If the faculty member chooses to refer the matter to the campus Dean for Student Services or designee for disposition, the Plenary Disciplinary Procedure shall be followed, and the student’s dismissal from the college is a possibility.

Educational Accessibility

Students who have documented, diagnosed disabilities, and who need special accommodations for tests, etc., are advised to see the Educational Accessibility Disabilities Services staff in Student Services so that the instructor may be notified of what accommodations are appropriate in each case. Requests for accommodations should be made to the designated campus Educational Accessibility counselor at least 45 days before classes begin. Documentation must be provided to support the need for accommodations.

For assistance with disabilities, contact the campus Educational Accessibility Counselor/Provider or the Coordinator of Educational Accessibility Services: call 757-822-7752, visit Student Services/Development, or visit the Educational Accessibility (URL provided in Important Websites section).

Emergency Procedures

In the event of a tornado or fire, students and staff may be directed to evacuate the building or move to an internal assembly area location within the building. If there is a bomb threat, shelter in place and wait for further directions from Emergency Responders. If told to evacuate, follow the route as outlined for Fire evacuation, unless directed out another direction by emergency personnel. Evacuation routes are posted in each classroom. The map indicates the route to the nearest exit. Students should review the map to make sure that the exit routes for the building are clearly understood. The information regarding locations of the Emergency Exterior Assembly Areas and Internal Assembly Areas for all classrooms or spaces used on the various campuses is available at the following link: [CEMP/IAA&EAA](#). If you will require assistance during an evacuation, let your instructor know at the end of the first class

Title 9

Title IX is a civil rights law that prohibits discrimination on the basis of sex in educational programs, activities, admission, and employment. Complaints of sex-based discrimination, sexual violence, domestic violence, dating violence, and sexual or gender-based harassment are governed by the college's Title IX Policy. To learn more about Title IX, sexual harassment, or sex-based discrimination, please review this [video](#). To report a Title IX violation, please visit the [Title IX web page](#)

Student Success Resources

The following resources are available to TCC students. See the *Student Handbook* for more information about student services and locations.

Library

A library is located at each TCC campus and at the Visual Arts Center. These libraries are intended for research and study, and they contain materials in print and digital format to support the courses, curricula, and mission of the college. The research materials include books, newspapers, magazines, journals, DVDs, streaming media and an extensive collection of indexes, abstracts and full-text databases. Faculty members may place materials on reserve in the libraries for their students. Visit this site for more information: <http://libguides.tcc.edu/LibraryPage>

Academic Support Services

Each campus provides various kinds of academic assistance. One-on-one tutoring, math and computer labs, and other forms of individual and group assistance may be available. Students can also find free help for writing, from short questions about commas and comma splices to a comprehensive review of research papers in progress, in the Writing Centers. Since I am home-based on the Chesapeake campus, then I strongly encourage choosing any of the Chesapeake tutors when prompted: <https://libguides.tcc.edu/tutoring>

Online Learning Help Desk

Visit the following website for Canvas support: <https://www.tcc.edu/student-services/canvas-support>

Canvas offers 24/7 support for students via phone (877-875-8359) and live chat. Live chat support can be accessed from the Help menu once you log into Canvas.

TCC provides Canvas support Monday to Friday, 8:30 a.m. to 5 p.m., by phone (757-822-1470), email (canvas@tcc.edu), and [online request form](#).

Online Help and Important Websites

- College Website: www.tcc.edu
- Closings and Emergencies: <http://www.tcc.edu/closings-emergencies>
- Student E-mail: <https://tcc.my.vccs.edu>
- Educational Accessibility: <http://www.tcc.edu/student-services/personal-support/students-disabilities>
- Chesapeake tutoring: <https://help.tcc.edu/s/article/Chesapeake-Campus-Tutoring>
- Student Handbook: <http://web.tcc.edu/student-handbook>
- TCC Catalog: <http://www.tcc.edu/academics/catalog/>
- Class Schedule: <https://m.sis.vccs.edu/index.php/app/catalog/classSearch?institution=TC295> (or log-in to SIS for current course offerings)
- Academic Calendar: <http://www.tcc.edu/academics/calendars/>
- Library: <https://libguides.tcc.edu/LibraryPage>.
- Financial aid information/assistance: <http://www.tcc.edu/paying-for-tcc/financial-aid/> or <http://studentaid.ed.gov/>